

§ 2—1

1. 设函数 $f(x)$ 在点 x_0 处可导,

$$\text{则 } \lim_{\Delta x \rightarrow 0} \frac{f[x_0 - 3(\Delta x)^2] - f(x_0)}{\sin^2 \Delta x} = (\text{ B }).$$

(A) $f'(x_0)$; (B) $-3f'(x_0)$;

(C) $3f'(x_0)$; (D) 0.

解: 由于 $f(x)$ 在点 x_0 处可导, 所以

$$\begin{aligned} & \lim_{\Delta x \rightarrow 0} \frac{f[x_0 - 3(\Delta x)^2] - f(x_0)}{\sin^2 \Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f[x_0 - 3(\Delta x)^2] - f(x_0)}{(\Delta x)^2} \\ &= -3 \lim_{\Delta x \rightarrow 0} \frac{f[x_0 - 3(\Delta x)^2] - f(x_0)}{-3(\Delta x)^2} \\ &= -3f'(x_0). \quad \text{选(B).} \end{aligned}$$

2. 函数

$$f(x) = \begin{cases} \sin x, & x \leq 0, \\ \sqrt{1+x} - \sqrt{1-x}, & 0 < x \leq 1 \end{cases}$$

在点 $x=0$ 处是 (A).

(A) 连续且可导;

(B) 不连续且不可导;

(C) 连续但不可导;

(D) 不连续但可导.

解: 因为

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \sin x = 0 = f(0), \\ \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (\sqrt{1+x} - \sqrt{1-x}) = 0 \\ &= f(0), \end{aligned}$$

故函数 $f(x)$ 在 $x=0$ 处连续.

$$\begin{aligned} \text{又由于 } & \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0^+} \frac{\sqrt{1+x} - \sqrt{1-x} - 0}{x} \end{aligned}$$

$$= \lim_{x \rightarrow 0^+} \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})}$$

$$= \lim_{x \rightarrow 0^+} \frac{2}{\sqrt{1+0} + \sqrt{1-0}} = 1,$$

极限存在, 所以函数

$f(x)$ 在 $x=0$ 处可导. 选 (A).

3. 确定常数 a, b , 使得函数

$$f(x) = \begin{cases} ax + b\sqrt{x}, & x > 1 \\ x^2, & x \leq 1 \end{cases}$$

在 $x=1$ 处可导.

解: $f(x)$ 在 $x=1$ 处可导的

必要条件是 $f(x)$ 在 $x=1$ 处连续,

$$\text{而 } f(1^-) = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1,$$

$$f(1^+) = \lim_{x \rightarrow 1^+} f(x)$$

$$= \lim_{x \rightarrow 1^+} (ax + b\sqrt{x}) = a + b,$$

$$\text{由 } f(1^-) = f(1^+),$$

可得 $a + b = 1$.

$$\text{又 } f'_-(1) = \lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1^-} \frac{x^2 - 1^2}{x - 1} = \lim_{x \rightarrow 1^+} (x + 1) = 2,$$

$$f'_+(1) = \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1^+} \frac{(ax + b\sqrt{x}) - 1^2}{x - 1}$$

$$= \lim_{x \rightarrow 1^+} \frac{(ax + b\sqrt{x}) - (a + b)}{x - 1}$$

$$= \lim_{x \rightarrow 1^+} \frac{a(x - 1) + b(\sqrt{x} - 1)}{x - 1}$$

$$= \lim_{x \rightarrow 1^+} \left(a + \frac{b}{\sqrt{x} + 1} \right) = a + \frac{1}{2}b,$$

$$\text{由 } f'_+(1) = f'_-(1) = 2,$$

$$\text{得 } a + \frac{1}{2}b = 2.$$

解得 $a = 3$, 则 $b = -2$.

§ 2—2

1. 设 $y = \sqrt{x}(x^3 - 4\cos x - \sin 1)$,

求 y' 及 $y'|_{x=1}$.

解: $y' = \frac{1}{2\sqrt{x}}(x^3 - 4\cos x - \sin 1)$
 $+ \sqrt{x}(3x^2 + 4\sin x),$

$$\begin{aligned} y'|_{x=1} &= \frac{1}{2\sqrt{1}}(1^3 - 4\cos 1 - \sin 1) \\ &+ \sqrt{1}(3 \cdot 1^2 + 4\sin 1) \\ &= \frac{1}{2}(1 - 4\cos 1 - \sin 1) + (3 + 4\sin 1) \\ &= \frac{7}{2} + \frac{7}{2}\sin 1 - 2\cos 1. \end{aligned}$$

2. 设 $f(x) = x(x-1)(x-2)\cdots(x-99)$,

求 $f'(0)$.

解: 由导数定义, 得

$$\begin{aligned} f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{x(x-1)(x-2)\cdots(x-99) - 0}{x} \\ &= \lim_{x \rightarrow 0} (x-1)(x-2)\cdots(x-99) \text{ (连续)} \\ &= (0-1)(0-2)(0-3)\cdots(0-99) \\ &= (-1)^{99} \cdot 99! = -99!. \end{aligned}$$

3. 求 $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$ 的导数.

解: $y' = \left(\sqrt{x + \sqrt{x + \sqrt{x}}} \right)'$
 $= \frac{1}{2\sqrt{x + \sqrt{x + \sqrt{x}}}} \left(x + \sqrt{x + \sqrt{x}} \right)'$

$$\begin{aligned}
&= \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}} \left[1 + \frac{(x+\sqrt{x})'}{2\sqrt{x+\sqrt{x}}} \right]' \\
&= \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}} \left(1 + \frac{1+\frac{1}{2\sqrt{x}}}{2\sqrt{x+\sqrt{x}}} \right) \\
&= \frac{4\sqrt{x+\sqrt{x}} \cdot \sqrt{x} + 2\sqrt{x} + 1}{8y \cdot \sqrt{x+\sqrt{x}} \cdot \sqrt{x}}.
\end{aligned}$$

4. 求过原点且与曲线 $y = \frac{x+9}{x+5}$ 相切的直线方程.

解: 设直线与曲线 $y = \frac{x+9}{x+5}$

在点 (x_0, y_0) 相切,

则过原点的直线方程可设为 $y = \frac{y_0}{x_0} x$.

由于直线与曲线 $y = \frac{x+9}{x+5}$

在点 (x_0, y_0) 相切,

$$\text{则 } y_0 = \frac{x_0+9}{x_0+5} = 1 + \frac{4}{x_0+5},$$

$$\text{又 } y' = \frac{x+5-(x+9)}{(x+5)^2} = \frac{-4}{(x+5)^2},$$

如果 $x_0 \neq 0$,

$$\text{则 } \frac{y_0}{x_0} = \frac{-4}{(x_0+5)^2},$$

$$\frac{x_0+9}{(x_0+5)x_0} = \frac{-4}{(x_0+5)^2},$$

$$\frac{x_0+9}{x_0} = \frac{-4}{x_0+5}$$

$$x_0^2 + 18x_0 + 45 = 0,$$

$$(x_0+3)(x_0+15) = 0,$$

解得 $x_0 = -3$ 或 $x_0 = -15$, 则相应地

$$y_0 = \frac{-3+9}{-3+5} = \frac{6}{2} = 3,$$

$$\text{或 } y_0 = \frac{-15+9}{-15+5} = \frac{-6}{-10} = \frac{3}{5},$$

则过点 $(-3, 3)$ 的切线方程为

$$y = \frac{-3}{3}x = -x,$$

则过点 $\left(-15, \frac{3}{5}\right)$ 的切线方程为

$$y = \frac{\frac{3}{5}}{-15}x = -\frac{1}{25}x,$$

如果 $x_0 = 0$, 直线上 $y_0 = 0$,

曲线上 $y_0 = \frac{0+9}{0+5} \neq 0$, 矛盾!

5. $y = \sqrt{x-x^2} + \arcsin \sqrt{x}$, 求 y' .

$$\text{解: } y' = \left(\sqrt{x-x^2}\right)' + \left(\arcsin \sqrt{x}\right)'$$

$$= \frac{1-2x}{2\sqrt{x-x^2}} + \frac{1}{2\sqrt{x}\sqrt{1-(\sqrt{x})^2}}$$

$$= \frac{1-2x}{2\sqrt{x-x^2}} + \frac{1}{2\sqrt{x}\sqrt{1-x}}$$

$$= \frac{1-2x+1}{2\sqrt{x}\sqrt{1-x}} = \frac{1-x}{\sqrt{x}\sqrt{1-x}}$$

$$= \sqrt{\frac{1-x}{x}}.$$

6. 设 $y = f(\ln \cos(e^x))$,

其中 $f'(u)$ 存在, 计算 y' .

$$\text{解: } y' = f'(\ln \cos(e^x)) \cdot [\ln \cos(e^x)]'$$

$$= f'(\ln \cos(e^x)) \cdot \frac{1}{\cos(e^x)} \cdot [-\sin(e^x)] \cdot e^x$$

$$= -e^x \cdot \tan(e^x) \cdot f'(\ln \cos(e^x)).$$

§ 2—3

1. 已知函数 $y = \ln(x + \sqrt{1+x^2})$, 求 y'' .

$$\text{解: } y' = \frac{1}{x + \sqrt{1+x^2}} \left(1 + \frac{x}{\sqrt{1+x^2}} \right)$$

$$= \frac{1}{\sqrt{1+x^2}},$$

$$y'' = -\frac{1}{2}(1+x^2)^{-\frac{3}{2}} \cdot 2x = -\frac{x}{\sqrt{(1+x^2)^3}}.$$

2. 设 $f(x) = \ln(1+x)$,

求 $f^{(n)}(x)$ 及 $f'''\left(\frac{1}{2}\right)$.

$$\text{解: } f'(x) = \frac{1}{1+x} \cdot (1+x)'$$

$$= \frac{1}{1+x} = (1+x)^{-1},$$

$$f''(x) = \left[(1+x)^{-1} \right]' = -(1+x)^{-2},$$

$$f'''(x) = -\left[(1+x)^{-2} \right]'$$

$$= (-1)(-2)(1+x)^{-3},$$

.....

$$f^{(n)}(x) = (-1)(-2)\cdots[-(n-1)](1+x)^{-n}$$

$$= (-1)^{n-1}(n-1)!(1+x)^{-n}.$$

$$f''\left(\frac{1}{2}\right) = (-1)^{3-1}(3-1)!\left(1+\frac{1}{2}\right)^{-3}$$

$$= 2!\left(\frac{3}{2}\right)^{-3} = 2\left(\frac{2}{3}\right)^3 = 2 \cdot \frac{8}{27} = \frac{16}{27}.$$

3. 设 $y = f^3(x) + f(e^x)$,

其中 $f(x)$ 二阶可导, 求 y'' .

$$\begin{aligned}
 \text{解: } y' &= 3f^2(x) \cdot f'(x) + e^x f'(e^x), \\
 y'' &= 6f(x) \cdot [f'(x)]^2 + 3f^2(x) \cdot f''(x) \\
 &\quad + e^x f'(e^x) + (e^x)^2 f''(e^x) \\
 &= 6f(x) \cdot [f'(x)]^2 + 3f^2(x) \cdot f''(x) \\
 &\quad + e^x f'(e^x) + e^{2x} f''(e^x).
 \end{aligned}$$

§ 2—4

$$1. \text{ 求由方程 } \arctan \frac{y}{x} = \ln \sqrt{x^2 + y^2}$$

所确定的隐函数 $y = y(x)$ 的导数.

解: 方程两边对 x 求导, 得

$$\begin{aligned}
 \frac{\left(\frac{y}{x}\right)'}{1 + \left(\frac{y}{x}\right)^2} &= \frac{1}{\sqrt{x^2 + y^2}} \left(\sqrt{x^2 + y^2}\right)', \\
 \frac{\frac{xy' - y}{x^2}}{1 + \left(\frac{y}{x}\right)^2} &= \frac{1}{\sqrt{x^2 + y^2} \cdot 2\sqrt{x^2 + y^2}} \cdot (x^2 + y^2)'
 \end{aligned}$$

$$\frac{xy' - y}{x^2 + y^2} = \frac{2x + 2y \cdot y'}{2(x^2 + y^2)},$$

$$\frac{xy' - y}{x^2 + y^2} = \frac{x + y \cdot y'}{x^2 + y^2},$$

$$xy' - y = x + y \cdot y'$$

$$\text{解得 } y' = \frac{x + y}{x - y}.$$

$$2. \text{ 设 } \begin{cases} x = 2t^3 + 2, \\ y = e^{2t}, \end{cases} \text{ 求 } \frac{dy}{dx}, \frac{d^2y}{dx^2}.$$

$$\text{解: } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2e^{2t}}{6t^2} = \frac{e^{2t}}{3t^2} = \frac{1}{3} e^{2t} t^{-2},$$

$$\frac{d^2 y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dx} = \frac{d\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

$$= \frac{\frac{1}{3}(2e^{2t}t^{-2} - 2e^{2t}t^{-3})}{6t^2}$$

(分子按乘积求导)

$$= \frac{e^{2t}(t-1)}{9t^5}.$$

2. 设 $y = (\sin x)^{\cos x} \quad \left(0 < x < \frac{\pi}{2}\right)$,

求 y' .

解: $y' = (e^{\cos x \ln \sin x})'$

$$= e^{\cos x \ln \sin x} (\cos x \ln \sin x)'$$

$$= (\sin x)^{\cos x} \left(-\sin x \cdot \ln \sin x + \cos x \cdot \frac{\cos x}{\sin x} \right)$$

$$= (\sin x)^{\cos x} \left(-\sin x \cdot \ln \sin x + \frac{\cos^2 x}{\sin x} \right).$$

解法二: 两边取对数, 得

$$\ln y = \cos x \ln \sin x,$$

上式两边对 x 求导, 得

$$\frac{1}{y} \cdot y' = -\sin x \cdot \ln \sin x + \cos x \cdot \frac{\cos x}{\sin x},$$

于是

$$y' = y \left(-\sin x \cdot \ln \sin x + \frac{\cos^2 x}{\sin x} \right)$$

$$= (\sin x)^{\cos x} \left(-\sin x \cdot \ln \sin x + \frac{\cos^2 x}{\sin x} \right).$$

4. 求 $y = (x-1)^{\frac{5}{3}} \left(\frac{x-2}{x-3} \right)^{\frac{1}{2}}$ 的导数.

解：两边取对数，得

$$\ln|y| = \frac{5}{3}\ln|x-1| + \frac{1}{2}\ln|x-2| - \frac{1}{2}\ln|x-3|,$$

上式两边对 x 求导，得

$$\frac{1}{y} \cdot y' = \frac{5}{3(x-1)} + \frac{1}{2(x-2)} - \frac{1}{2(x-3)},$$

于是

$$y' = y \left[\frac{5}{3(x-1)} + \frac{1}{2(x-2)} - \frac{1}{2(x-3)} \right].$$

§ 2—5

1. 设函数 $y = f(x)$ 在点 x_0 处可导，

$$\Delta y = f(x_0 + \Delta x) - f(x_0),$$

则当 $\Delta x \rightarrow 0$ 时，必有 (). (B)

(A) dy 是 Δx 的等价无穷小量；

(B) $\Delta y - dy$ 是比 Δx 高阶的无穷小量；

(C) dy 是比 Δx 高阶的无穷小量；

(D) $\Delta y - dy$ 是 Δx 的同阶无穷小量.

解：因为

$$\Delta y = f(x_0 + \Delta x) - f(x_0) = dy + o(\Delta x)$$

$$(\Delta x \rightarrow 0),$$

所以 $\Delta y - dy = o(\Delta x) \quad (\Delta x \rightarrow 0)$.

$$\text{则 } \lim_{\Delta x \rightarrow 0} \frac{\Delta y - dy}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{o(\Delta x)}{\Delta x} = 0.$$

选(B).

2. 设 $y = f(\ln x) + \ln f(x)$,

其中 $f(x)$ 二阶可导，

则 $dy = ($). (D)

$$(A) \frac{f'(\ln x)}{x} dx; \quad (B) \frac{f'(x) + xf'(x)}{\ln x} dx;$$

$$(C) \frac{f(x)f'(x) + xf'(\ln x)}{xf(x)} dx;$$

$$(D) \frac{f(x)f'(\ln x) + xf'(x)}{xf(x)} dx.$$

$$\text{解: } \frac{dy}{dx} = f'(\ln x) \cdot (\ln x)' + \frac{f'(x)}{f(x)}$$

$$= \frac{1}{x} f'(\ln x) + \frac{f'(x)}{f(x)}$$

$$= \frac{f(x)f'(\ln x) + xf'(x)}{xf(x)}, \text{ 则}$$

$$dy = y'dx = \frac{f(x)f'(\ln x) + xf'(x)}{xf(x)} dx.$$

故应选 (C).

$$\text{补充: } \frac{d^2y}{dx^2} = -\frac{1}{x^2} f'(\ln x) + \frac{1}{x} f''(\ln x) \cdot \frac{1}{x}$$

$$+ \frac{f''(x)f(x) - [f'(x)]^2}{f^2(x)}$$

$$= \frac{1}{x^2} [f''(\ln x) - f'(\ln x)]$$

$$+ \frac{f''(x)f(x) - [f'(x)]^2}{f^2(x)}.$$

3. 以下命题正确的是 (). (A)

(A) 如果 $f(x)$ 在点 x_0 处可导,

则 $f(x)$ 在点 x_0 处连续;

(B) 如果 $f(x)$ 在点 x_0 处连续,

则 $f(x)$ 在点 x_0 处可导;

(C) 如果 $f(x)$ 在点 x_0 处连续,

则 $f(x)$ 在点 x_0 处可微;

(D) 如果 $f(x)$ 在点 x_0 处可导,

则 $f(x)$ 在点 x_0 处不一定可微.

4. 设 $y = \ln \sqrt{1-x^2}$, 求 dy .

解: $y = \frac{1}{2} \ln(1-x^2)$,

$$y' = \frac{-2x}{2(1-x^2)} = -\frac{x}{1-x^2},$$

所以 $dy = y'dx = -\frac{x}{1-x^2} dx$.

5. 设方程 $y = x + \arccos y$

确定了函数 $y = y(x)$, 求 dy .

解: 方程两边求微分, 得

$$dy = dx - \frac{1}{\sqrt{1-y^2}} dy,$$

$$\text{得} \quad \left(1 + \frac{1}{\sqrt{1-y^2}}\right) dy = dx,$$

$$\text{解得} \quad dy = \frac{\sqrt{1-y^2}}{1+\sqrt{1-y^2}} dx.$$

解法二: 方程两边关于 x 求导, 得

$$y' = 1 - \frac{1}{\sqrt{1-y^2}} y',$$

$$\text{整理得} \quad y' = \frac{\sqrt{1-y^2}}{1+\sqrt{1-y^2}}. \text{ 于是}$$

$$dy = y'dx = \left(\frac{\sqrt{1-y^2}}{1+\sqrt{1-y^2}} \right) dx.$$